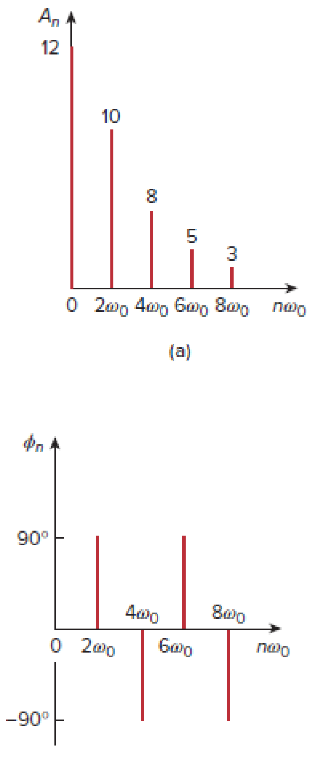
Problem

The amplitude and phase spectra of a truncated Fourier series are shown in Fig. 17.85.

(a) Find an expression for the periodic voltage using the amplitude-phase form. See Eq. (17.10).

(b) Is the voltage an odd or even function of *t*?

Figure 17.85:



Eq. (17.10):

$$\boxed{f(t) = a_0 + \sum_{n=1}^{\infty} A_n \cos(n\omega_0 t + \phi_n)} \qquad \textbf{17.10}$$

Step-by-step solution

Step 1 of 3

Consider the following phase spectra:

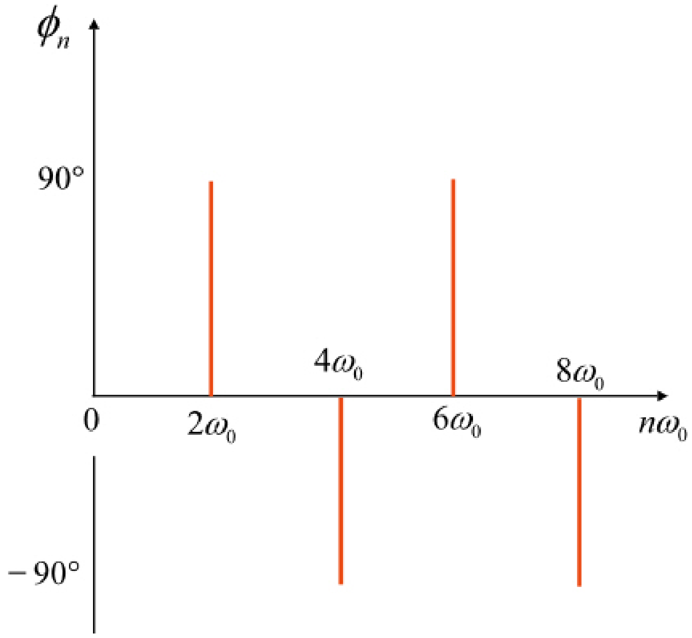


Figure 2

Step 2 of 3

(a)

Write the expression for amplitude-phase form.

$$f(t) = a_0 + \sum_{n=1}^{\infty} A_n \cos(n\omega_0 t + \phi_n)$$

Expand the above expression.

$$f(t) = a_0 + A_1 \cos(\omega_0 t + \phi_1) + A_2 \cos(2\omega_0 t + \phi_2) + \ldots + A_8 \cos(8\omega_0 t + \phi_8)$$

From Figures (1) and (2),

$$f(t) = \begin{Bmatrix} a_0 + A_2 \cos(2\omega_0 t + \phi_2) + A_4 \cos(4\omega_0 t + \phi_4) + \\ A_6 \cos(6\omega_0 t + \phi_6) + A_8 \cos(8\omega_0 t + \phi_8) \end{Bmatrix} \quad \ldots\ldots (1)$$

Where,

$A_2, A_4, A_6,$ and A_8 are the amplitudes.

$\phi_2, \phi_4, \phi_6,$ and ϕ_8 are the phase angles.

Write the expression for periodic voltage using expression (1).

$$v(t) = \begin{Bmatrix} 12 + 10\cos(2\omega_0 t + 90°) + 8\cos(4\omega_0 t - 90°) + \\ 5\cos(6\omega_0 t + 90°) + 3\cos(8\omega_0 t - 90°) \end{Bmatrix}$$

Therefore, the expression for periodic voltage is

$$\boxed{\begin{Bmatrix} 12 + 10\cos(2\omega_0 t + 90°) + 8\cos(4\omega_0 t - 90°) + \\ 5\cos(6\omega_0 t + 90°) + 3\cos(8\omega_0 t - 90°) \end{Bmatrix}}.$$

Step 3 of 3

(b)

If the plot of $v(t)$ is symmetrical about the vertical axis it is said to be even otherwise it is said to be odd. That is,

$$v(t)=v(-t)$$

Even function contains cosine terms where as odd function contains sine terms. Therefore, it is clear that from the expression of the $v(t)$ in part (a), $v(t)$ contains the cosine terms, and then it is said to be even.

Therefore, the $\boxed{v(t) \text{ is an even function of } t}$.